

worksheet-mcq-generator_1.1 Introduction - Revisiting Number Systems

Generation 1

Tool: Worksheet & MCQ Generator

Class: Class 10

Subject: Maths

Topic: Chapter 1: Real Numbers

Subtopic: 1.1 Introduction - Revisiting Number Systems

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Chapter 1: Real Numbers

Sub Topic: 1.1 Introduction - Revisiting Number Systems

Overview

In this section, we will revisit the number systems that you have learned so far, including natural numbers, whole numbers, integers, rational numbers, and irrational numbers. We will also explore how these different types of numbers relate to each other and form the set of real numbers.

Worksheet: Revisiting Number Systems

Multiple Choice Questions (MCQs)

1. Which of the following is not a natural number?

- A) 1
- B) 5
- C) 0
- D) 10

Answer: C) 0

Explanation: Natural numbers start from 1 and go up (1, 2, 3, ...).

1. Which of the following represents a whole number?

- A) -2
- B) 0.5
- C) 0
- D) 1.5

Answer: C) 0

Explanation: Whole numbers include all natural numbers plus zero (0, 1, 2, 3, ...).

True or False Questions

1. All integers are rational numbers.

- **Answer:** True

Explanation: Rational numbers can be expressed as a ratio of two integers, and since integers themselves can be written as fractions (e.g., $3 = 3/1$), they are rational.

1. Every rational number is an integer.

- **Answer:** False

Explanation: While all integers are rational numbers, not all rational numbers are integers (e.g., $1/2$ is a rational number but not an integer).

Fill in the Blanks

1. The set of natural numbers is denoted by _____.

- **Answer:** N

Explanation: The symbol 'N' is commonly used to denote the set of natural numbers.

1. The set of whole numbers is denoted by _____.

- **Answer:** W

Explanation: The symbol 'W' is commonly used to denote the set of whole numbers.

Short Answer Questions

1. List three examples of irrational numbers.

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- **Answer:** $\sqrt{2}$, π , e

Explanation: These numbers cannot be expressed as a simple fraction and have non-repeating, non-terminating decimal expansions.

1. What is the difference between a rational number and an irrational number?

- **Answer:** A rational number can be expressed as a ratio of two integers, while an irrational number cannot be expressed as such a ratio and has a non-repeating, non-terminating decimal expansion.

Explanation: This highlights the fundamental difference based on their representation and decimal properties.

Problem Solving

1. If x is a rational number and y is an irrational number, what can be said about $x + y$?

- **Answer:** $x + y$ is irrational.

Explanation: Adding a rational number to an irrational number results in an irrational number because the sum cannot be expressed as a simple fraction.

1. Simplify the expression $\sqrt{16}$.

- **Answer:** 4

Explanation: The square root of 16 is 4, which is a natural number.

Application Questions

1. Find the smallest natural number that is divisible by both 4 and 5.

- **Answer:** 20

Explanation: The least common multiple (LCM) of 4 and 5 is 20, which is the smallest number divisible by both.

1. Determine if the number 0.333... (repeating) is rational or irrational.

- **Answer:** Rational

Explanation: 0.333... can be expressed as the fraction $\frac{1}{3}$, making it a rational number.

Advanced Questions

1. Prove that the sum of two irrational numbers can be rational.

- **Answer:** Consider $\sqrt{2}$ and $-\sqrt{2}$. Both are irrational, but their sum is $\sqrt{2} + (-\sqrt{2}) = 0$, which is rational.

Answer: Consider $\sqrt{2}$ and $-\sqrt{2}$. Both are irrational, but their sum is 0, which is rational.

Explanation: This demonstrates that the sum of two irrational numbers can indeed be rational.

1. Is the product of two irrational numbers always irrational? Justify your answer.

• **Answer:** No, it is not always irrational. For example, $\sqrt{2} \times \sqrt{2} = 2$, which is rational.

Explanation: This shows that the product of two irrational numbers can sometimes result in a rational number.

Critical Thinking

1. Can every real number be represented on a number line? Explain.

• **Answer:** Yes, every real number can be represented on a number line.

Explanation: The number line includes all points corresponding to real numbers, whether they are rational or irrational.

1. Why is the set of real numbers considered complete?

• **Answer:** The set of real numbers is considered complete because it contains all possible limits of sequences of rational numbers, including both rational and irrational numbers.

Explanation: This completeness property means there are no "gaps" in the real number system.

Conceptual Understanding

1. How do rational and irrational numbers differ in terms of their decimal representations?

• **Answer:** Rational numbers have either terminating or repeating decimal representations, whereas irrational numbers have non-repeating, non-terminating decimal representations.

Explanation: This distinction helps in identifying whether a given number is rational or irrational based on its decimal form.

1. What is the significance of the number π in mathematics?

• **Answer:** π is significant because it represents the ratio of a circle's circumference to its diameter and is an irrational number.

Explanation: π is crucial in geometry and trigonometry due to its role in calculations involving circles and spheres.

Reflection

1. Reflect on why understanding the classification of numbers is important in mathematics.

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- **Answer:** Understanding the classification of numbers helps in solving equations, performing operations, and understanding the properties of numbers, which are foundational for advanced mathematical concepts.

Explanation: This reflection emphasizes the importance of number classification in building a strong mathematical foundation.

1. How does the concept of real numbers unify the various number systems you have studied?

- **Answer:** The concept of real numbers unifies natural numbers, whole numbers, integers, rational numbers, and irrational numbers into a single, comprehensive system that encompasses all possible numerical values.

Explanation: This unification allows mathematicians to work within a consistent framework that includes all types of numbers encountered in practical and theoretical contexts.

This worksheet covers a range of question types to ensure a thorough understanding of revisiting number systems in the context of real numbers.